

Shivam Jadhav

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SUMMARY

Customer-obsessed and impact-driven **Software Engineer & Meta Certified Front-End Developer** with 5+ years of experience delivering scalable, secure, and user-focused solutions across cloud, web, and AI systems. Proficient in full-stack (MERN), backend (Node.js), cloud-native (AWS, Docker, Kubernetes), and AI/ML technologies. Adept at designing microservices, building CI/CD pipelines, automating testing, and translating complex needs into elegant, production-ready software.

EDUCATION & RESEARCH

Master's in Computer Science | University of North Texas, Denton, USA

Expected May 2025

- **Relevant Coursework:** Software Engineering, Special Problems, Artificial Intelligence, Computer Architecture
- **Research:** **Telemetry Data Simulation** – Designed a secure IoT communication framework using microservices, TCP/IP, socket programming, and raw data handling for seamless data exchange with a central server.

SKILLS

- **Full - Stack & Frontend:** Expert in MERN stack (MongoDB, Express.js, React.js, Node.js), Hooks, Redux, Tailwind, and REST/GraphQL APIs.
- **Backend & Microservices:** Proficient in Java, Node.js, Express, SQL/NoSQL, Redis, JWT, API versioning, and scalable service design.
- **Cloud & DevOps:** Advanced AWS user (EC2, Lambda, API Gateway, S3, DynamoDB, ELB, CloudWatch, Route 53). Skilled in Docker, Docker Compose, Kubernetes, CI/CD, Linux, and systemd.
- **AI & LLMs:** Applied NLP and transformer models (BERT, RoBERTa, ALBERT) using PyTorch and Hugging Face; experience with CUDA and AI model deployment.
- **Testing & Automation:** Experience with Jest, Mocha, Cypress, Postman, and test-driven development (TDD) practices.
- **System Engineering:** Expertise in server imaging (RAID), NVIDIA GPU setups, bash scripting, and performance monitoring with Grafana.
- **Design Thinking:** Skilled in Figma, wireframing, semantic HTML, accessibility, SEO best practices, and responsive UI/UX design.
- **Foundations:** Strong grasp of data structures, algorithms, graph theory, problem-solving, and system architecture design.
- **Version Control & Collaboration:** Git/GitHub, GitLab, Agile/Scrum, documentation writing, and collaborative software development.

EXPERIENCE

IT Specialist - Student Assistant | University of North Texas, Denton, USA

Sep 2023 – Present

- **Spearheaded** the design and deployment of a Universal Digital Signage System using React.js, Node.js, MongoDB, and Docker on Raspberry Pi, reducing manual setup time by 90% and cutting university costs by \$40,000.
- **Engineered** a system-monitoring solution (Bash, SSH, Grafana) that proactively tracked uptime and sent real-time alerts, improving server reliability and reducing incident response time by 60%.
- **Managed** end-to-end infrastructure: RAID imaging (Ubuntu 24), NVIDIA CUDA setup, full server rack maintenance, and performance visualization, ensuring zero downtime for AI research.
- **Demonstrated strong collaboration**, problem-solving, and adaptability in cross-functional teams delivering production-grade backend tools and scripts.

Full-Stack Engineer | Cyrrup Solutions, Hyderabad, India

Mar 2021 – Aug 2023

- **Led development** of an IoT fleet management platform using Node.js, Vue.js, Redis, and microservices, enabling scalable control over 40+ devices.
- **Optimized cloud systems** (AWS EC2, ELB, CloudWatch) and automated deployments using Docker and GitLab CI/CD, reducing latency and manual errors.
- **Collaborated** with DevOps, designers, and testers to ensure highly-available, fault-tolerant services using test-driven development and Agile workflows.
- Resolved critical system issues (DNS, TCP/IP, container networking), improving uptime by 30%.

Freelance Full-Stack Developer | ArkTech Ltd.

Jan 2019 – Apr 2021

- **Architected and delivered** Ark CRM for finance sector clients using MERN stack, streamlining customer workflows for 10K+ active users.
- **Built AI-powered recommendations** using Python, scikit-learn, cosine similarity, and custom scraping logic, increasing user engagement by 25%.

- **Practiced secure coding**, authentication (JWT), and scalable API design, while collaborating remotely with clients and project stakeholders.

PROFESSIONAL PROJECTS

Universal Digital Signage System | University of North Texas

Sep 2023 – Present

- **Led the full lifecycle** of a scalable digital signage platform — from Figma wireframes, user personas, and system architecture to production deployment — serving 50+ Raspberry Pi devices across campus.
- **Designed and implemented** the complete stack: responsive React.js frontend, modular Node.js backend, and a clustered MongoDB database for high availability and horizontal scalability.
- **Architected the system** using microservices, with clean API contracts, CDN integration for optimized media delivery, and strict separation of frontend/backend concerns via Docker Compose environments (dev/prod).
- **Built a custom CI/CD pipeline** with shell scripting and GitHub Actions, deploying the full tech stack (frontend, backend, MongoDB, NGINX) with a single Bash command; reduced deployment time from 1 hour to under 5 minutes.
- **Configured production servers** with RAID for fault tolerance, NGINX for reverse proxying, and SSL-secured domain linking, ensuring a robust, reliable infrastructure.
- **Applied software engineering best practices:** SOLID principles, RESTful API design, JWT/OAuth2 authentication, semantic HTML, accessibility, logging, and error handling.
- **Demonstrated leadership** through cross-functional collaboration, GitHub issue tracking, code reviews, branching strategies, and effective resolution of production issues, while mentoring peers and maintaining documentation.
- **Project page:** divisionbyinfinity.github.io/posts/uds/

Vehicle Black Box (Fleet Management and AI Solution)

- **Designed a real-time fleet tracking system** with Vue.js, Node.js, Redis, and Python, using MQTT and Web-Socket for live data exchange.
- **Leveraged AWS and GCP services** (S3, Pub/Sub, Cloud Functions) for elastic scaling and efficient event-driven processing.
- **Followed clean architecture**, using separation of concerns and modular services to increase testability and reduce coupling.

RESEARCH

AI Bug Priority Problem | MS-CS Research

Aug 2024 – Dec 2024

- **Conducted research** under Prof. Wajdi Aljedani on prioritizing software bugs using NLP, aiming to improve accessibility issue resolution in large codebases.
- **Trained transformer models** (BERT, RoBERTa, ALBERT) using PyTorch and Hugging Face, achieving up to **96% accuracy**.
- **Deployed AI models and plugin tools** using Streamlit and Render for practical demonstration.
- Live demo:  [AI Model](#)  [Web Plugin](#)

Railway Bogie Defect Detection using AI and IoT

- **Built a real-time defect detection system** for railway components using PyTorch, scikit-learn, and live camera feeds.
- **Integrated end-to-end pipeline** using React, Tailwind, Firebase, and Nodemailer for defect alerting and device control.
- **Demonstrated AI robustness** across lighting variations, achieving high accuracy and field applicability.

CERTIFICATIONS & AWARDS

- **Meta Certified Front-End Developer** – [Coursera](#), [Credly](#)
- **AWS Certified Solutions Architect – Associate (SAA-C03)** – *Expected June 2025*
- **Amazon Web Services Cloud Practitioner** – [Udemy](#)
- **Advanced React by Meta** – [Coursera](#)
- **Google: Tools of the Trade: Linux and SQL** – [Coursera](#)
- Meta HTML and CSS in Depth – [Coursera](#)
- Programming with JavaScript – [Coursera](#)
- Version Control – [Coursera](#)
- Docker for Developers – [LinkedIn Learning](#)
- Deep Learning – [Coursera](#)

REFERENCES

- **Michael Baggett**
Assistant Dean of Information Technology, University of North Texas
Email: michael.baggett@unt.edu
Recommendation: *Available on LinkedIn*